ETH zürich ALMA MATER STUDIORUM UNIVERSITÀ DI BOLOGNA

CachePool: Many-core cluster of customizable, lightweight scalar-vector PEs for irregular L2 data-plane workloads

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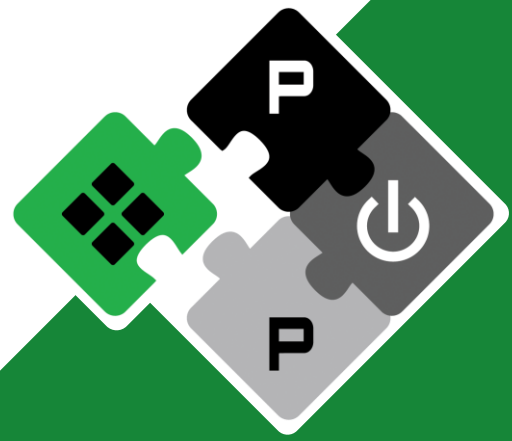

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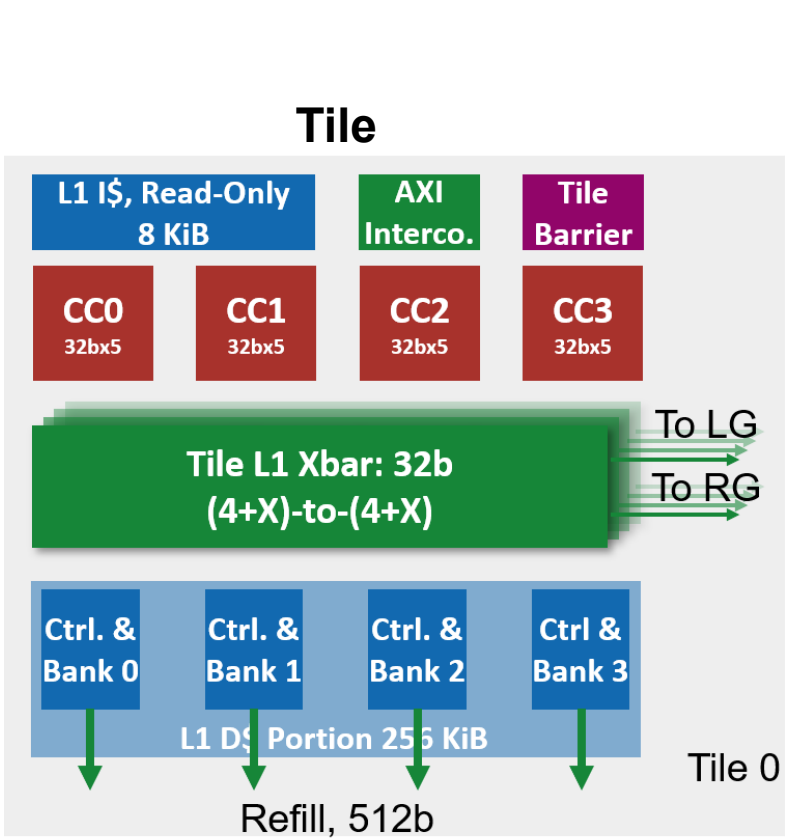
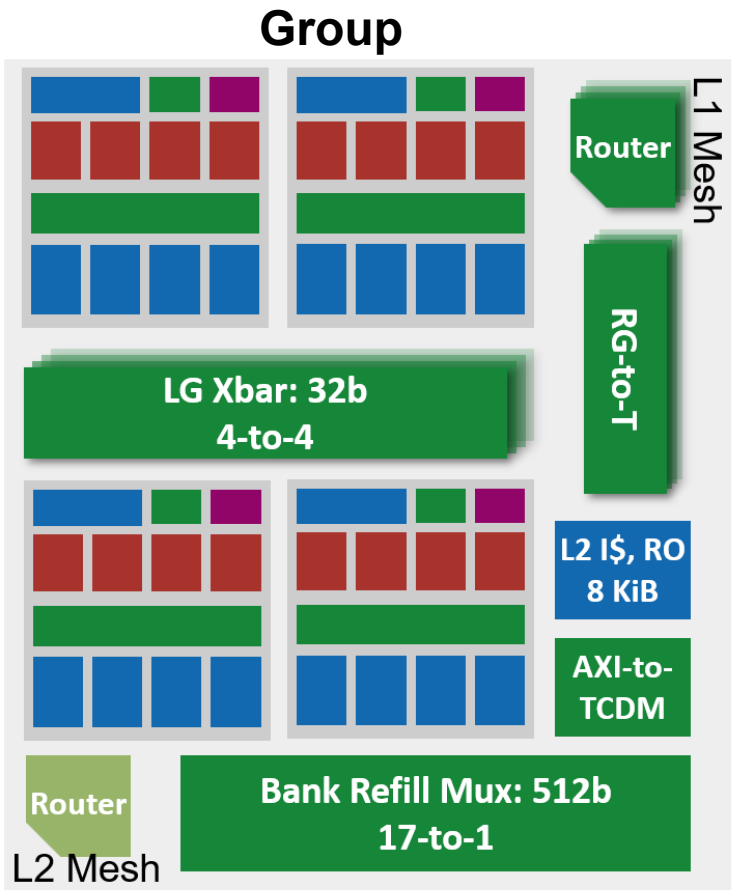
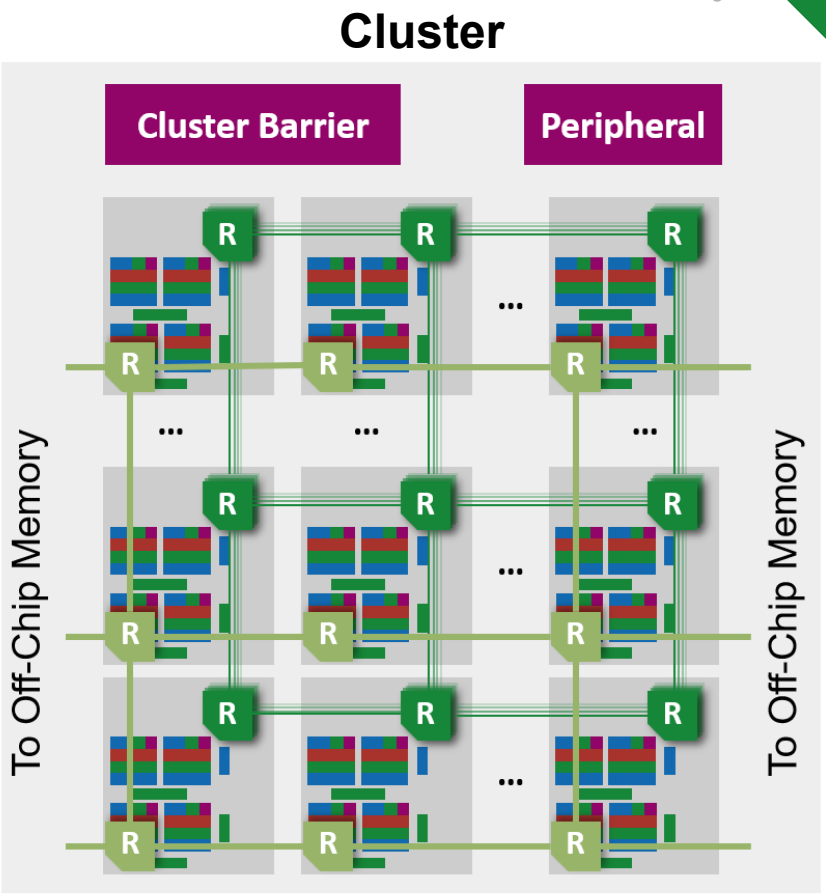
PULP Platform
Open Source Hardware, the way it should be!

Hardware Development (Cluster)

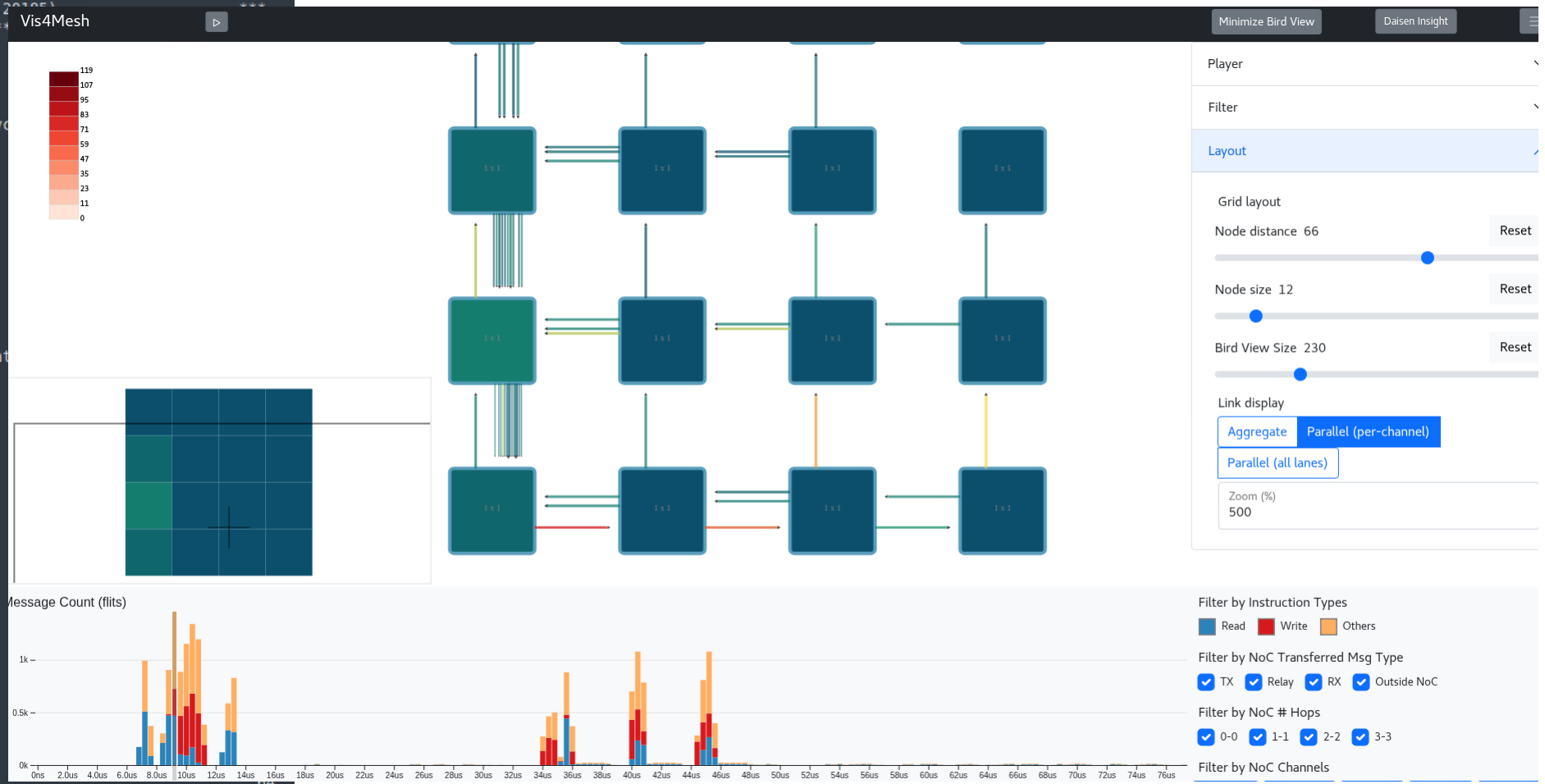




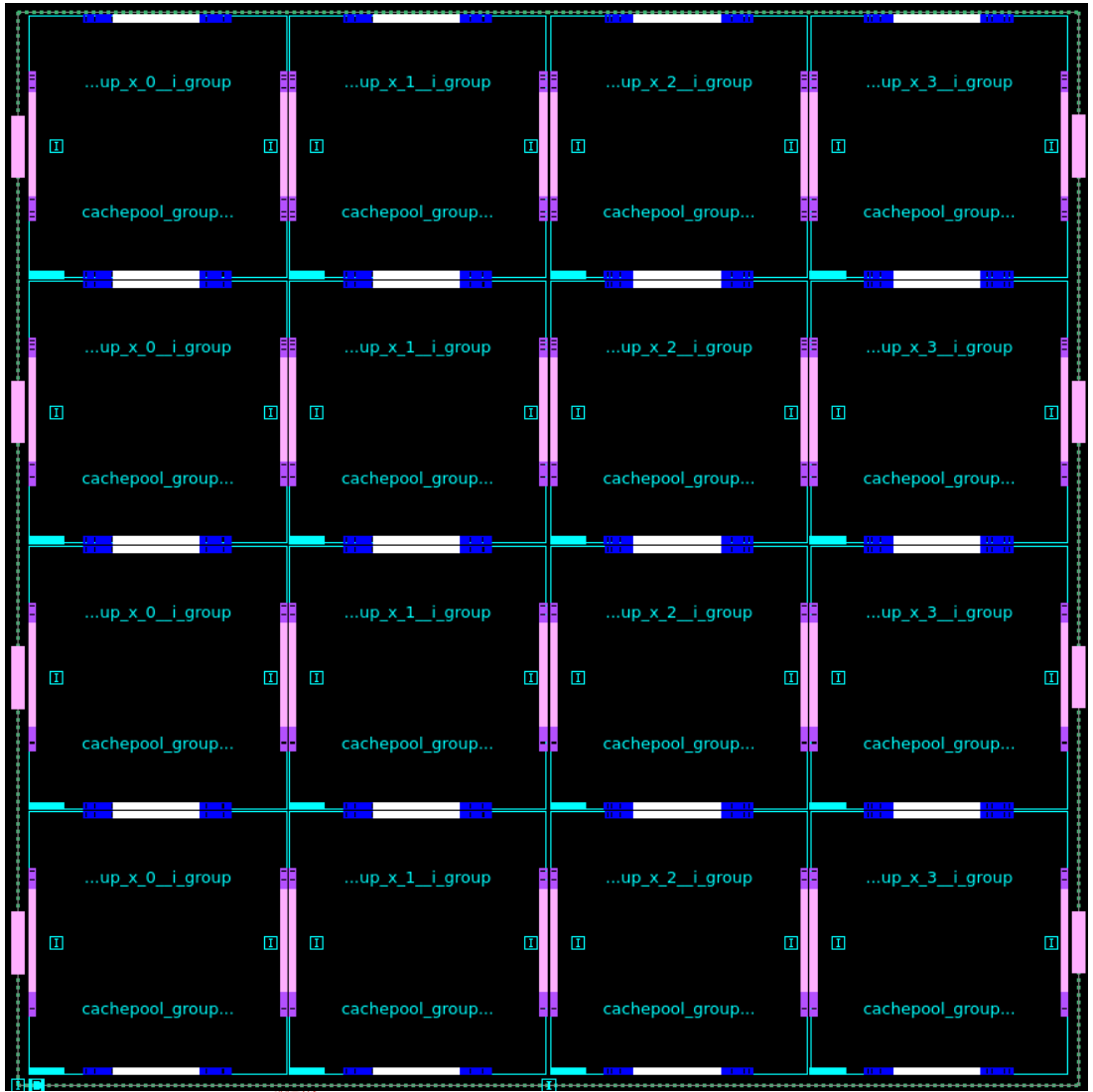


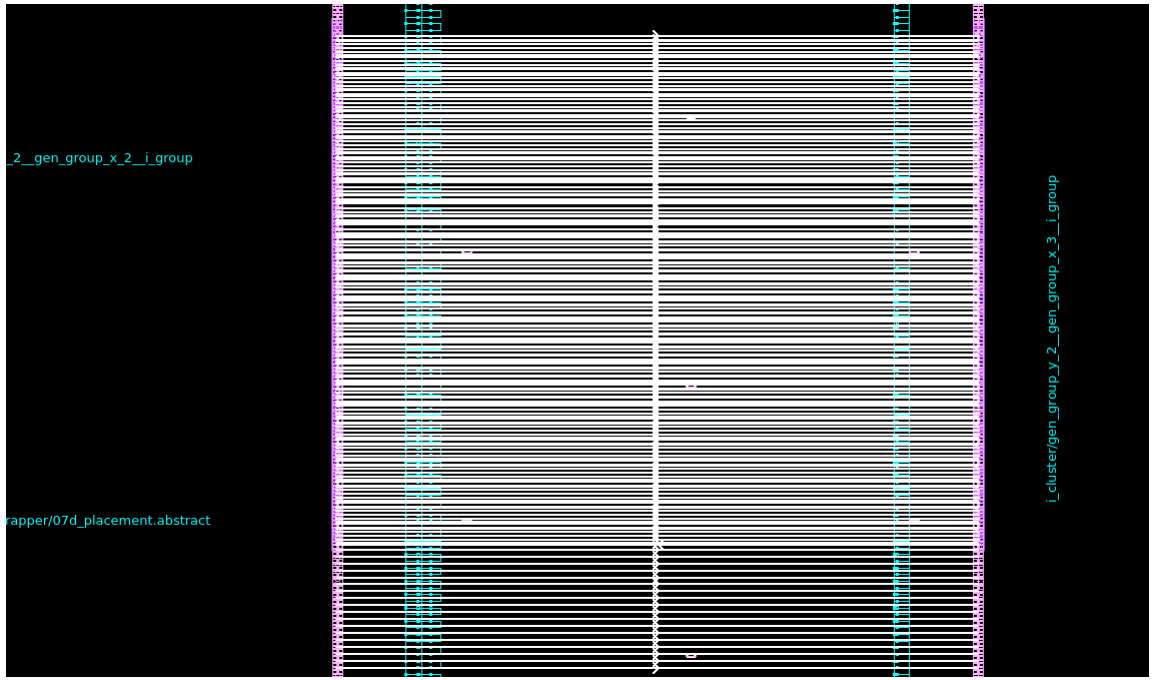
Hardware Development (Cluster)

```
**********************************************************************
*** Session 3 (kernel): cycles [19475, 20185)                      ***
**********************************************************************
***           Snitch Scalar Core
   Inst Fetch Accepted Count:
   Inst Fetch AVG Serving Cycles:
   Scalar LSU Load Req-to-Commit Count:
   Scalar LSU Load AVG Req-to-Commit Cyc
   Stall Cycles Total:
   Stall Cycles (No Valid Instr):
   Stall Cycles (ITLB Miss):
   Stall Cycles (RAW Hazard):
   Stall Cycles (LSU Port Busy):
   Stall Cycles (Accelerator Port):
   Stall Cycles (Fence Drain):

   VLSU Load Req-to-Commit Count:
   VLSU Load AVG Req-to-Commit Cyc:

***           Spatz Controller Utilizat
   VLSU:
   VLSU Active (not accurate):
   VLSU Num Instructions:
   VLSU Stall Cycles:
   VLSU WR Stalls:
   VFU:
   VFU Active (not accurate):
   VFU Num Instructions:
   VFU Stall Cycles:
   VFU WR Stalls:
   VSLDU:
   VSLDU Active (not accurate):
   VSLDU Num Instructions:
   VSLDU Stall Cycles:
   VSLDU WR Stalls:

***           Spatz VLSU Utilization
   Number of VRF Valid Cycles:
   Number of VRF Transaction Counts:
   VRF W Utilization:
   VRF W AVG Cycles:
   Number of Mem Valid Cycles:
   Number of Mem Transaction Counts:                    64
   Mem Utilization:                                 100.00
```



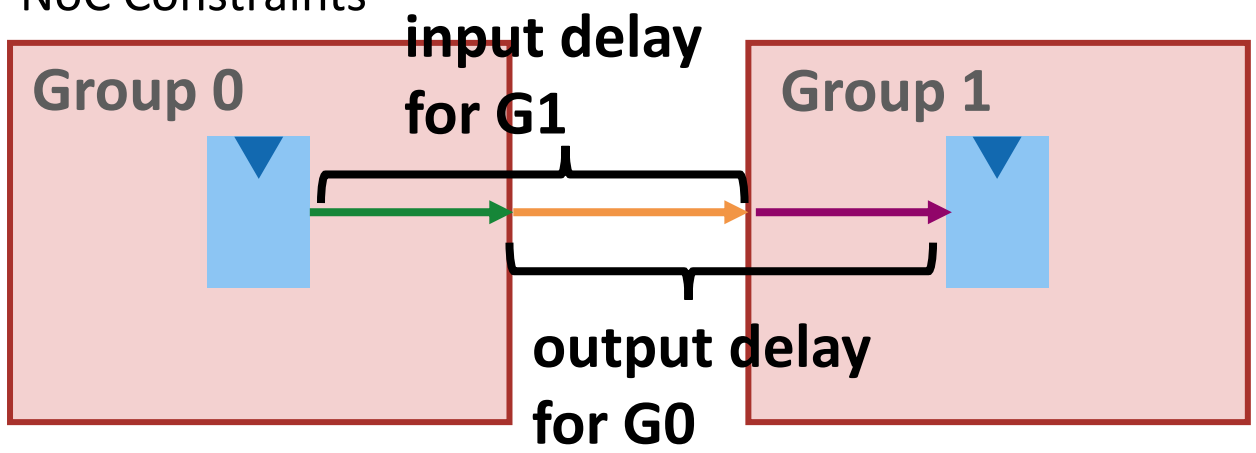
Physical Design (Multi-Group)





Physical Design (Multi-Group)

- **Group-Level**
  - NoC Constraints



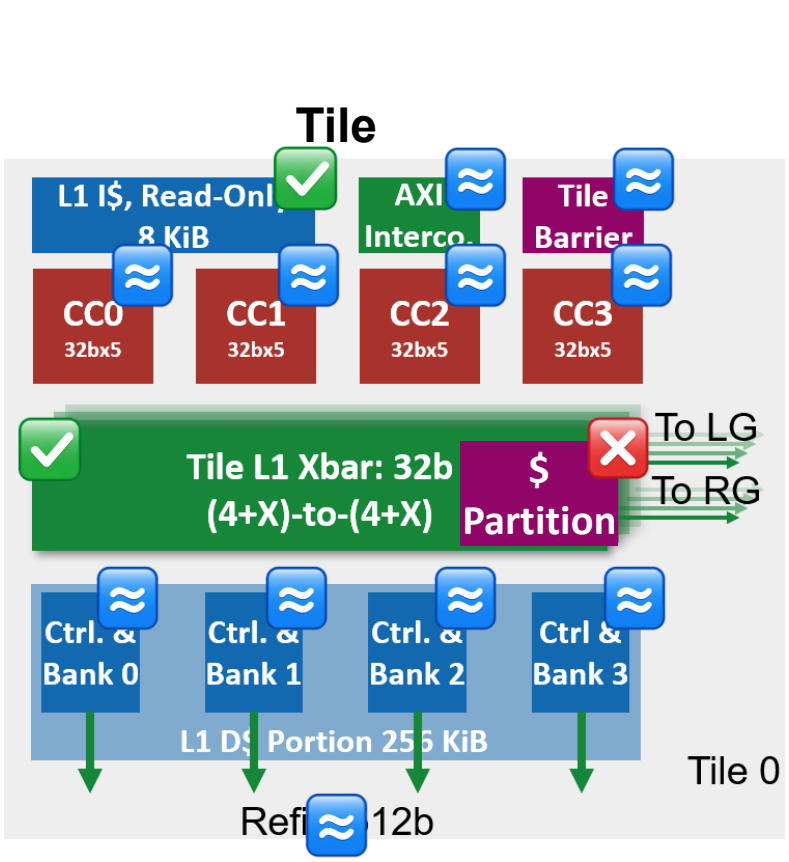
reg2out

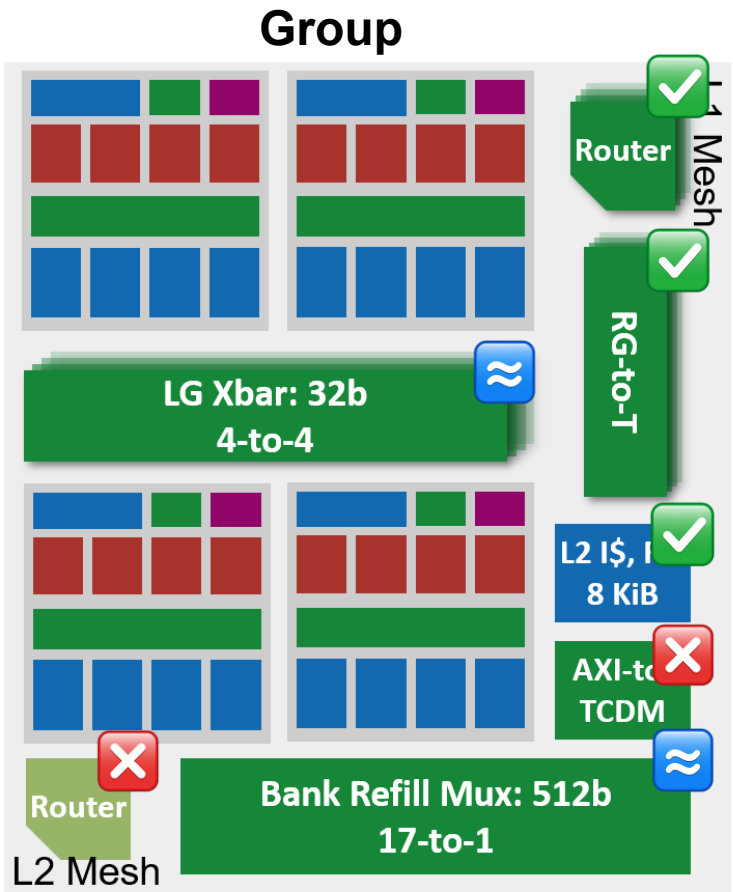
feedthrough (cluster)

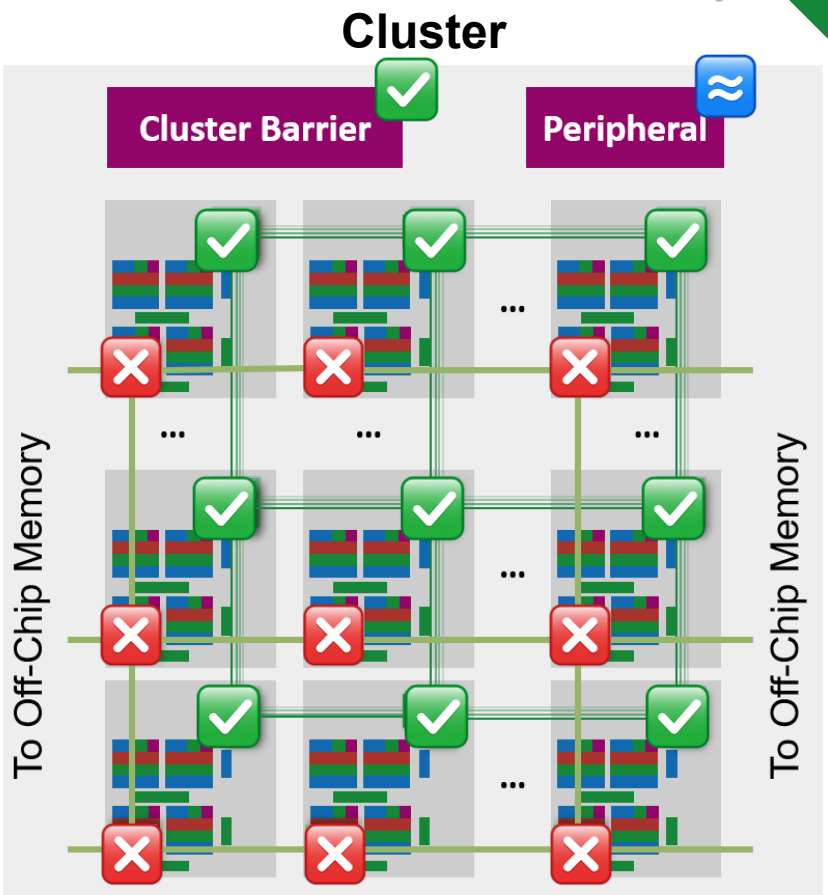
in2reg

  - 1st iteration shows some timing issues on peripheral paths
    - Applied cuts onto them (not performance critical
  - 2nd iteration under going
- **Cluster-Level**
  - FP, PG and pin placement done
  - Now waiting for group level to continue placement (need IO timing info)

GVSoC Modeling Progress







Done, the same logic as RTL

In place, with approximate model

Not implemented yet

GVSoC Modeling Progress

- **Model bugs fixed**
  - **Barrier and AMO**
    - The hardware barrier was mapped to the wrong address, so it never blocked and broke cross-core sum
    - The atomic write resp result was written to the wrong buffer, cause spin-lock livelock
  - **Private stack**: All 16 cores shared one stack, overwriting each other's frames (crash + broke prints)
  - **Cache and memory access**
    - The vector unit marked memory bursts "done" at issue time, so consumers could read data before it actually arrived → commit bursts at issue + real latency.
    - A second reader of a line still being refilled got an instant fake "hit" on a not-installed line (early data race) → keep the line "pending" and make followers wait for the refill.
    - The backing memory had zero latency, so every cache miss was nearly free → use DRAMsys

GVSoC Modeling Progress

- **Results**
  - Now all 8 CI test kernels are passed:
    - fdotp-32b_M8192 ✅ pass → ✅ pass
    - fdotp-32b_M32768 ✅ pass → ✅ pass
    - gemv-opt_M512_N128_K32 ✅ pass → ✅ pass
    - fmatmul-32b_M32_N32_K32 ✅ pass → ✅ pass
    - fft-32b_M1024_N16 ❌ wrong result → ✅ pass
    - load-store_M16 ❌ hang → ✅ pass
    - RLC_M1_N1350_K10 ❌ fail → ✅ pass
    - byte-enable ❌ fail → ✅ pass
    - spin-lock ❌ livelock → ✅ pass
  - Performance calibration is ongoing

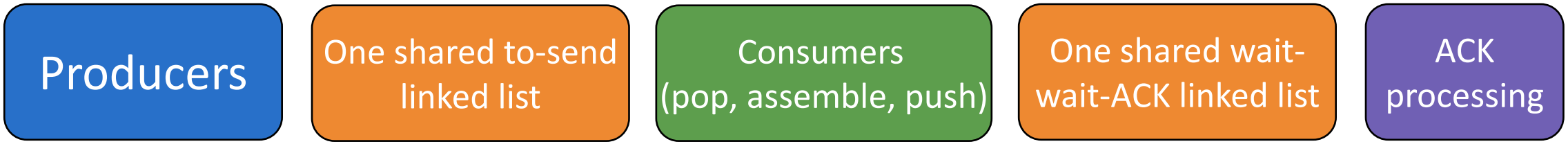
RLC kernel multi-user use cases dev

- The kernel used to have **one** RLC entity (one UE) with one global state struct; now it has an **array of N entities** and every packet is routed to **its** entity

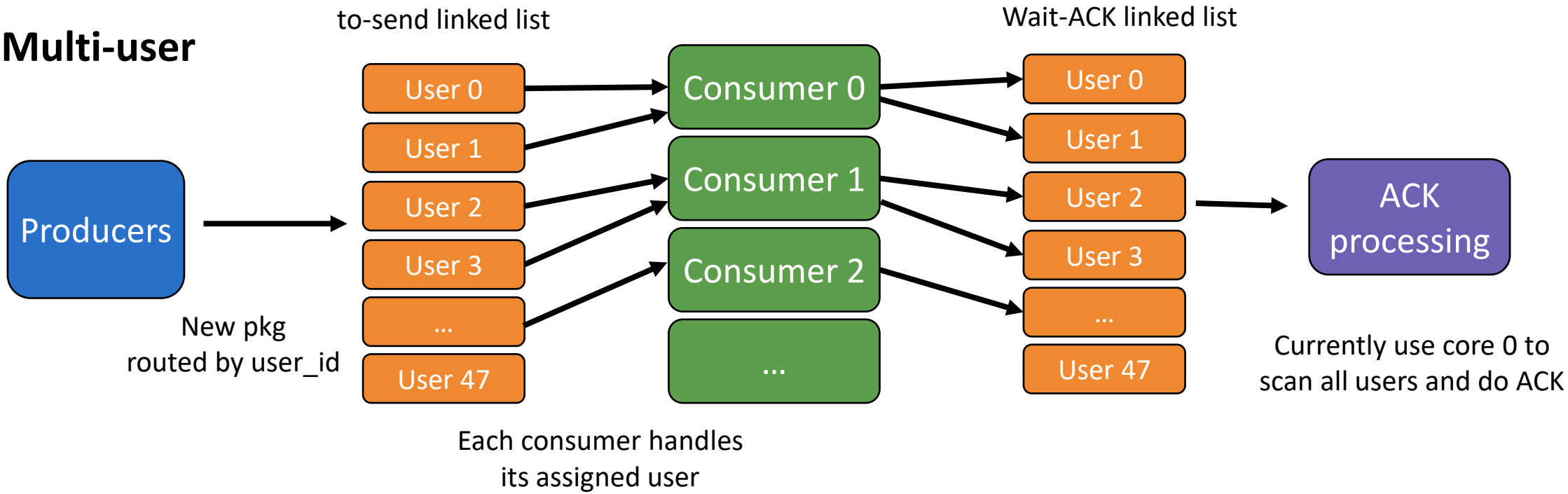
Component	BEFORE — simgle user only	AFTER — N users
RLC context	One global *rlc_ctx* containing both queues, SN state, polling state, and statistics	*rlc_ctx*[NUM_USERS]: one independent context per UE
List locks	Two global locks shared by all cores	Per-user to-send and wait-ACK locks, distributing contention
Sequence numbers	One global *vtNext* sequence-number stream	Independent *rlc_ctx[u].vtNext* stream for each RLC entity
Packet routing	Descriptor *user_id* exists but is ignored	user_id selects the target RLC context and queue
User count	One user	Compile-time NUM_USERS, derived from ACTIVE_USER_NUMBER

RLC kernel flow change

Single-user



Multi-user



RLC kernel multi-user use cases dev

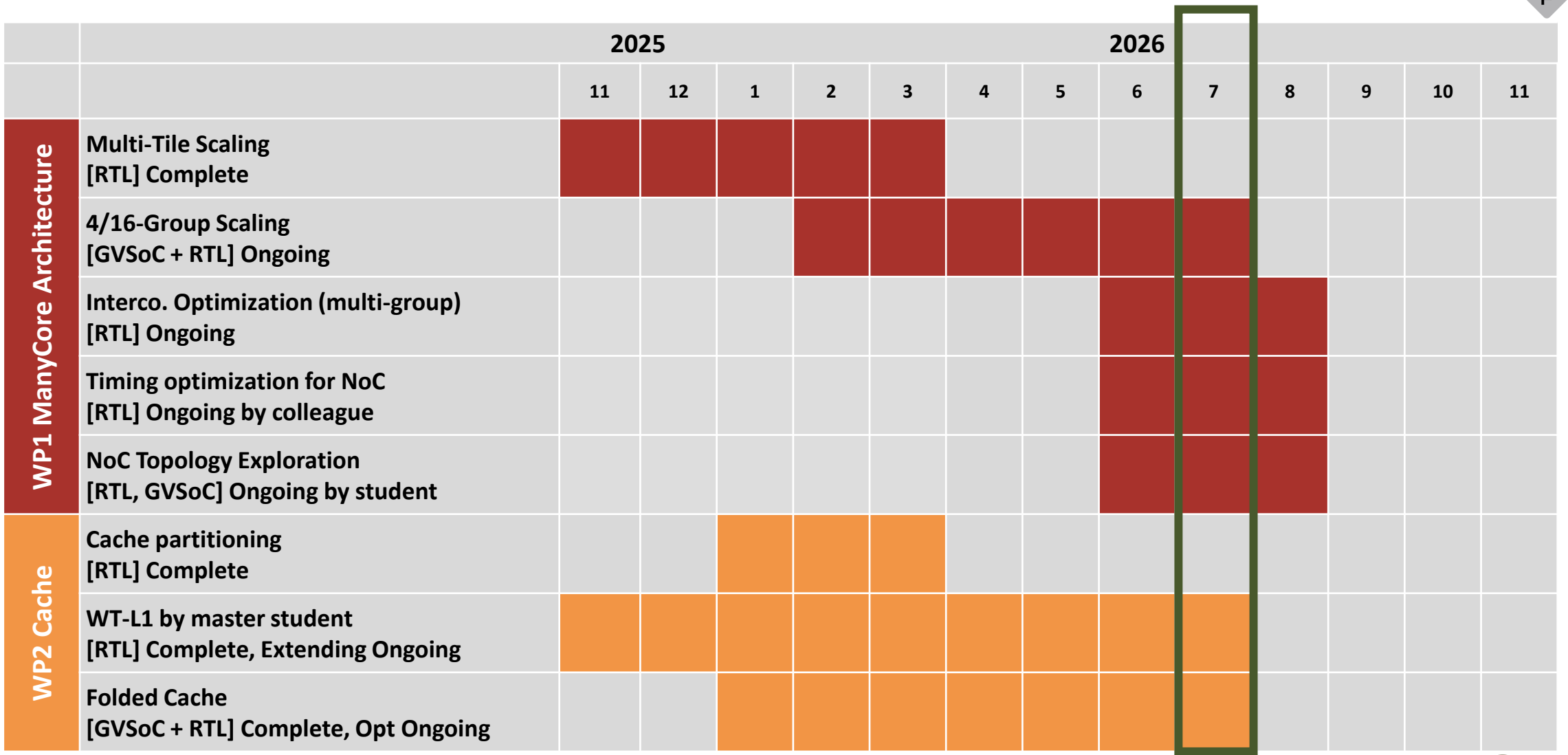
- Tested run in **GVSoC** model and **RTL** with small hw config
  - Single group hardware config, 16 cores
  - 48 users, 300pkg, 800B PDUs
  - GVSoC: P2/C2, P4/C4, P2/C8, P4/C8 pass
    - Next step is run with larger config, and report and compare the performance number
  - RTL: P2/C2,P4/C4, P2/C8 pass; P4/C8 fails
    - Need to debug the rtl simulation

	Kernel run cycle	Kernel run cycle		
SW config	GVSoC	RTL	Diff	Throughput (Gbps)
P2/C2	538635	530059	1.6%	3.56
P2/C8	505146	457988	10.3%	3.80
P4/C4	301139	316688	-4.9%	6.38
P4/C8	250210	fail		7.67

Next Step

- **Diyou Side**
  - Physical design exploration for multi-group cluster => help to determine the BW we can have
    - First PD run looks very positive => we can increase BW between groups
  - Rebase onto main (folded cache work) and merge
  - Multi-Snitch cores sharing one Spatz
    - Plan
      Add additional Snitch into the current CC (what we have done in a different project)
    - Some needed additions
      1) Snitch inside the tile should all connect to the Spatz, using muxing to select the host
      2) Software and runtime support to reduce the coding complexity (some helper functions)
- **Zexin Side**
  - GVSoC development and calibration under multi-group config
  - SW development for multi-user, debug under multi-group config

Timeline


2025
2026
11
12
1
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3
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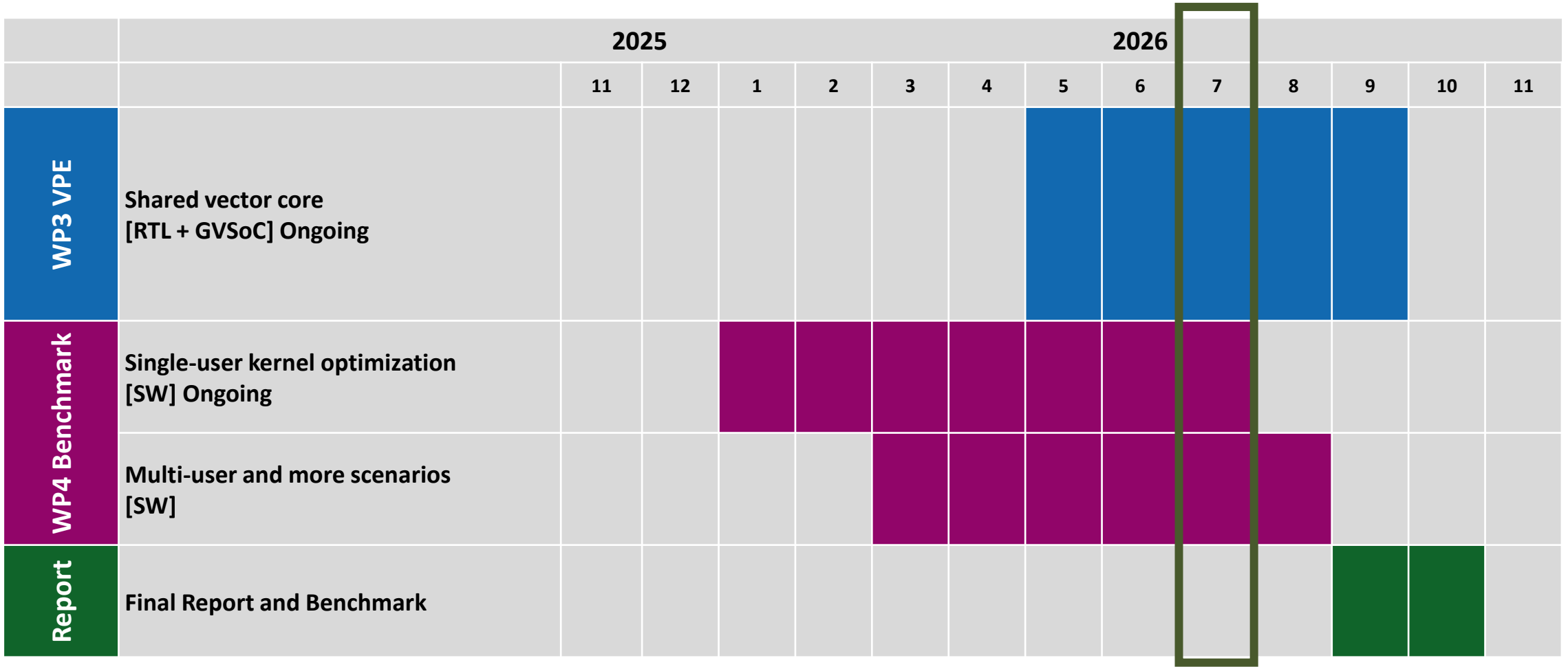
6
7
8
9
10
11
WP1 ManyCore Architecture
Multi-Tile Scaling
[RTL] Complete
4/16-Group Scaling
[GVSoC + RTL] Ongoing
Interco. Optimization (multi-group)
[RTL] Ongoing
Timing optimization for NoC
[RTL] Ongoing by colleague
NoC Topology Exploration
[RTL, GVSoC] Ongoing by student
WP2 Cache
Cache partitioning
[RTL] Complete
WT-L1 by master student
[RTL] Complete, Extending Ongoing
Folded Cache
[GVSoC + RTL] Complete, Opt Ongoing

Timeline


2025
2026
11
12
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6
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8
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10
11
WP3 VPE
Shared vector core
[RTL + GVSoC] Ongoing
WP4 Benchmark
Single-user kernel optimization
[SW] Ongoing
Multi-user and more scenarios
[SW]
Report
Final Report and Benchmark

L1 Access Latency

- **Problem: increased L1 access latency when scaling up**
- **Solutions**
  - **(Ongoing, RTL & Physical) Hardware optimization**
    Further optimize the interconnection and cache controller to reduce access latency on existing architecture
    => Retiming the current interconnection, remove unnecessary cuts in interco and cache
    => Target to reduce 1-2 cycles (~5 cyc) for hit-to-use latency
    Currently has reduced to 6 cycles:
    Core => Xbar => AMO => Ctrl (2) => AMO => Core
  - **(Complete, RTL) Cache partitioning**
    Partition the shared L1 to keep data localized to avoid remote access latency
  - **(Complete with some problems, RTL & Physical) WT-L1**
    Explore and evaluate a private WT L1 for each core
    => Performance degrade on single tile
    => Needs more work to support larger system

L1 Access Latency

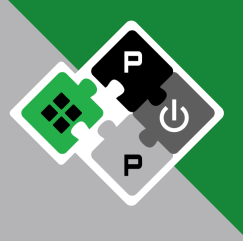
- **Problem: large NoC latency (~28 cyc based on previous exploration)**
- **Solutions**
  - **(Ongoing, Physical) Reduce hop latency**
    Explore the timing under 7nm technology to see the possibility of reducing per-hop latency to 1 cycle
    => Reduce the latency by 1/2
  - **(Planned, Physical) Cut clk-domain (optional)**
    If 1-cyc is not possible, explore to operate the NoC under a higher frequency than cores (e.g. 1.5GHz)
    => Effectively cut the latency by 1/3
  - **(Planned, GVSoC & Physical) NoC Topology**
    Explore different NoC topologies
    => Torus to reduce the diameter of NoC by 1/2, reduce longest latency by 1/2

PE-Related

- **Problem: Improve scalar core performance**
- **Solutions**
  - **(Planned, RTL & GVSoC) Shared vector core**
    Explore two Snitch sharing one Spatz configuration to improve scalar performance
    Software complexity needs to be considered, especially on the possible conflicting use of register file.
    Discussed Solution: critical section

RLC Workload

- **Problem: Current RLC model cannot reveal all properties of the kernel**
- **Solutions**
  - **(Ongoing, SW) Kernel optimization**
    Adapt the current model to better represent the real cases, e.g. do some operations on the data. Further cooperation with HQ side to develop a more complete and better model.
  - **(Planned, SW) Kernel development**
    Implement the missing use cases for multi-user



Thank you!